Guided Inquiry: Kinetics in a Plug Flow Reactor

Objective

In this activity, you will use the **PFR Kinetics Simulator** to explore how reaction rate expressions, stoichiometry, and inlet concentrations affect the performance of a Plug Flow Reactor.

Chemical Reaction:

$$NO_2 + CO \rightarrow NO + CO_2$$

Part 1: Elementary Reactions

Goal: Understand what makes a reaction "elementary" and how it relates to the rate law.

1. **Set the conditions:**
   - Set **Order w.r.t NO_2** to **1.0**.
   - Set **Order w.r.t CO** to **1.0**.
   - These exponents match the stoichiometric coefficients in the chemical equation ($1NO_2 + 1CO$).
2. **Observe:**
   - Look at the red line (NO_2) and the blue line (CO).
   - How does the rate of disappearance (the slope of the line) change as you move from left to right (increasing volume)?
   - *Question:* Why does the slope get flatter as volume increases?
3. **Definition:** When the exponents in the rate law match the stoichiometric coefficients exactly, the reaction is often called **elementary**. This implies the reaction happens in a single collision step exactly as written.

Part 2: The Chameleon Constant (k)

Goal: Discover why the units of the rate constant k must change when the rate law changes.

1. **Prediction:** The rate of reaction, r, always has units of **concentration / time** (e.g., mol/L/s).
2. **Experiment:**
   - Change the exponents using the sliders and observe the "Unit Display" under the formula box.
   - Fill in the table below:

Order (NO_2)	Order (CO)	Total Order (n)	Units of k
0	0	0	
1	0	1	
1	1	2	
1.5	0.5	2	

3. **Analysis:**
   - Compare the units for the two cases where the Total Order is 2. Are they the same or different?
   - *Challenge:* Write a general formula for the units of k in terms of concentration (M), time (s), and total order (n).

Part 3: Stoichiometric Consistency

Goal: Verify that changing the *speed* of the reaction does not change the *ratio* in which chemicals react.

1. **Set the conditions:**
   - Set Inlet $[NO_2]$ to **1.0 mol/L**.
   - Set Inlet $[CO]$ to **2.0 mol/L**.
   - Set Exponents to any value you like (try high values like 2.0, then low values like 0.5).
2. **Observe the gap:**
   - Look at the vertical distance between the Blue line (CO) and the Red line (NO_2).
   - Does this vertical distance change as you move down the reactor volume?
   - Does this vertical distance change when you make the reaction faster or slower using the exponent sliders?
3. **Conclusion:**
   - The reaction consumes 1 mole of NO_2 for every 1 mole of CO.
   - Therefore, the difference $[CO] - [NO_2]$ should always be equal to the difference in their initial concentrations: $2.0 - 1.0 = 1.0$.
   - *Principle:* Kinetics determines **how fast** we get there; Stoichiometry determines **where** we go.

Part 4: The Power of Concentration

Goal: Visualize how initial concentration drives the reaction rate.

1. **Set the conditions:**
   - Set **Order w.r.t NO_2** to **2.0**.
   - Set **Order w.r.t CO** to **0.0**.
   - (This effectively makes the rate law: $r = k[NO_2]^2$).
2. **Experiment:**
   - Start with Inlet $[NO_2] = 1.0$. Look at the slope of the red line at the very beginning (Volume = 0). It should be moderately steep.
   - Increase Inlet $[NO_2]$ to **2.0** (double the concentration).
3. **Analyze the Rate:**
   - Look at the new slope at Volume = 0. It is much steeper.

- Mathematically, if $r = k[\mathrm{NO_2}]^2$, and you double the concentration ($2x$), by what factor does the initial rate increase?
  a) $2x$
  b) $4x$
  c) $8x$
- Verify this visually by observing how quickly the reactant crashes towards zero compared to the first case.